Kathmandu University

Assignments I

Group – M. Sc. Subject : MATH 509

Text Book: Introduction to Real Analysis (4th Edition)

-By R.G Bartle and D.R. Sherbert

A

Define the following;

Cartesian product, function, Inverse Functions, absolute value, supremum, infimum, bounded function, neighborhood

Theorems, Examples and Lemma

a. Theorem 1.1.4 b. Theorem 1.3.13 c. Algebraic Properties 2.1.1 d. Theorem 2.1.7 e. Theorem 2.2.2 f. Example 2.2.6(a,b,c) g. Lemma 2.3.4 h. Theorem 2.4.8

B

Define the following;

Sequence of real number, converge and divergent sequence, tail sequence, bounded sequence, monotone sequence, subsequence, Cauchy sequence, series

Theorems, Examples and Lemma

a. Theorem 3.1.4 b. Theorem 3.1.5 c. Theorem 3.1.9 d. Theorem 3.2.2 e. Theorem 3.4.8 f. Lemma 3.5.3 g. Lemma 3.5.4 h. Cauchy Convergence Criterion 3.5.5 i. The nth Term Test 3.7.3

C

Define the following;

Cluster point, limit of f at c, bounded function, right-hand limit of f at c, continuous function**,** Continuous, absolute maximum, uniformly continuous, Lipschitz function

Theorems, Examples and Lemma

a. Theorem 4.1.5 b. Theorem 4.2.2 c. Theorem 4.2.4 d. Theorem 4.2.6 e. Theorem 4.2.7 f. Theorem 5.2.1 g. Theorem 5.2.6

D

1. Define the derivative and prove that differentiable function is continuous and converse may not be true.
2. State and prove the following theorems
   - Max min (extreme value) theorem
   - Local extreme theorem
   - Darboux's theorem
   - Rolle's Theorem
   - Mean value theorem
   - Increase and decrease theorem
   - Taylor's theorem
   - L' hospital's Rule.